Leeward Community College

Math and Science Division
Mechatronics Program

Course Syllabus

Course:	MECH191
CRN:	81145
Title:	Metallurgy

Course Description

This course provides a comprehensive introduction to the principles of metallurgy and material science, focusing on the structure, properties, and performance of metals and alloys. Students will explore the physical and chemical properties of various metals, the processes used to alter these properties, and their applications in manufacturing. Topics include mechanical testing, metallography, phase diagrams, heat treatment, mechanical properties of metals, corrosion, and failure analysis. Students will gain practical experience in material testing, alloy composition, and microstructural analysis as well as how to select and apply materials for various industrial purposes.

Course Dates and Time

Credits:	3
Semester:	Fall 2026
Start date:	August 24, 2026
End date:	December 18, 2026
Time:	1700-1930 (5:00 PM to 7:30 PM)
Classroom:	PS101B
Office Hours:	By Appointment Only
Prerequisites:	None

Contact Information	Instructor	Program Coordinator	Division
Name:	Kainani Santos	William “Bill” Labby	N/A
Office location:	CE401	CE401	BS 106A
Email:	kainanis@hawaii.edu	wlabby@hawaii.edu	N/A
Phone number:	(808)600-8362	(808)330-2175	(808)455-0251

Course Learning Objectives

1. Explain the relationship between the structure, properties, and performance of different metals and alloys.
2. Describe the phases in a metal alloy system using a phase diagram.
3. Apply knowledge of heat treatment processes to modify the properties of metal for specific engineering applications.
4. Conduct laboratory tests to measure mechanical properties such as hardness, tensile strength, and ductility.
5. Interpret failure mechanisms in metals, such as corrosion and fatigue, based on properties and usage conditions.
6. Assess the suitability of different metals and alloys for given engineering or manufacturing scenarios, considering factors such as performance, durability, and cost.

Course Information

Textbook/Reading Resources

Instructor will provide all materials in a digital format (i.e., PDF, video links). Students are responsible for making their own accommodations to modify the format of the content as needed (i.e., creating hardcopy printout of PDF).

Content is taken from:

- Metallurgy, 5th edition by B.J. Moniz
- Manufacturing Processes for Engineering Materials, 6th Edition by S. Kalpakjian

Technology Requirements

Students must have reliable access to communication tools such as their university-provided Gmail account, the learning management system, Lamaku, and access to the internet outside of the designated course meeting times. Students should also plan to bring their own resources such as a laptop. Issues with accessibility will be handled on a case-by-case matter.

Lecture and Demonstrations

This course is primarily a lecture-based class. There will be live and in-person demonstrations of various processes, as available. Demonstrations require that all participants wear proper PPE as designated by the instructor in advance of the meeting. Failure to have the proper PPE during demonstrations will result in the participant being asked to leave the demonstration area. Demonstrations will not be repeated for participants who are unavailable to attend. Accommodations will be handled on a case-by-case matter and be requested from the student to the instructor with as much notice as possible.

Grading Policy

Grades for the course will be calculated using the assessment rubric provided in this syllabus.

Assignment	**Total Per Course**	**Value per Assignment**	**Total Points**
Attendance	15	1	15
Metallography Portfolio	1	50	50
Unit Quizzes	5	5	25
Final Exam	1	10	10
			Total Points: 100

The Final Exam will be held at PS101A on **December 18, 2026**. The exam will begin at 1700 (5:00 PM) and must be returned to the instructor by 1900 (07:00 PM) to be considered for grading. Failure to return the exam by the ending time will result in the final exam score automatically being zero (0).

Grade Scale

Final letter grades are assigned based on the standard A-F grading scale.

Metallography Portfolio Modules
Students will complete one metallography module per instructional week using the shared class specimen kit (AmScope ME580). Completed modules are submitted as a single portfolio at the end of Week 17.

Acceptable Use of Artificial Intelligence for Coursework
While the use of Artificial Intelligence, such as ChatGPT, can offer students guidance, it is essential to uphold academic integrity and allow students to be evaluated based on their own understanding of course topics. Furthermore, students are expected to meet all course learning objectives and adhere to the parameters outlined within this syllabus.

In this course, students are not permitted to use any form of generative AI (i.e., ChatGPT, Claude, Microsoft Copilot, or Apple Intelligence) in whole or in part, to generate course materials or assignments. Usage of grammar and spell checking tools (i.e.,Grammarly) or those integrated into processing software (i.e., Microsoft Word, Google Docs) may be used to enhance the communication of ideas by student.

Additional Notifications for Students

Academic Dishonesty

Academic dishonesty cannot be condoned by the University. Such dishonesty includes cheating and plagiarism (examples of which are given below), which violate the Student Conduct Code and could result in expulsion from the University.

Cheating includes but is not limited to giving unauthorized help during an examination, obtaining unauthorized information about an examination before it is administered, using inappropriate sources of information during an examination, altering the record of any grades, altering answers after an examination has been submitted, falsifying any official University record, and misrepresenting the facts in order to obtain exemptions from course requirements.

Plagiarism includes but is not limited to submitting any document, to satisfy an academic requirement, that has been copied in whole or part from another individual's work without identifying that individual; neglecting to identify as a quotation a documented idea that has not been assimilated into the student's language and style, or paraphrasing a passage so closely that the reader is misled as to the source; submitting the same written or oral material in more than one course without obtaining authorization from the instructors involved; or dry-labbing, which includes (a) obtaining and using experimental data from other students without the express consent of the instructor, (b) utilizing experimental data and laboratory write-ups from other sections of the course or from previous terms during which the course was conducted, and (c) fabricating data to fit the expected results.

Please see the Leeward CC website for more information about academic dishonesty.

Participation Verification

Students who do not participate in their classes during the first week of the semester are considered to be "No Shows." The College assumes that "No Show" students no longer wish to participate in their education. Students who are identified as "No Shows" will be administratively disenrolled from their classes and given a 100% tuition refund. Participant Verification helps to release students who registered but did not intend to come to class from a financial obligation and failing grade. It also keeps the College in compliance with federal financial aid guidelines.

See Executive Policy 7.209 Student Participation Verification in Coursework.

Title IX

Leeward Community College is committed to supporting students and upholding the College's non-discrimination policy. Under Title IX, discrimination based upon sex and gender is prohibited. If you experience an incident of sex- or gender-based discrimination, harassment, or violence, we encourage you to reach out for help. While you may talk to a faculty member, understand that as a "Responsible Employee" of the College, the faculty member must inform the college's Title IX Coordinator to ensure that our students are supported and aware of the resources available. If you would like to speak with someone who can afford you with privacy and confidentiality, there are designated confidential resources available who can meet with you.

For more information about available resources, and University policies, please see our Title IX website or contact our Title IX Coordinator, Thomas Hirsbrunner at 808-455-0478 or email leetix@hawaii.edu.

Course Schedule*

Week	Date	Topics/Assignments
1	August 24, 2026	Syllabus, PPE Unit 1 – Foundations: What Metals Are and How They're Built
2	August 31, 2026	Unit 1 – Continued Unit 1 Quiz
3	September 7, 2026	NO CLASS
4	September 14, 2026	Unit 2 – Metal Families: Knowing What You're Working With
5	September 21, 2026	Unit 2 (Continued)
6	September 28, 2026	Unit 2 (Continued) Unit Quiz 2
7	October 5, 2026	Unit 3 – Testing, Inspection, and Quality: The Technician's Core Skills
8	October 12, 2026	Unit 3 (Continued)
9	October 19, 2026	Unit 3 (Continued) Unit 3 Quiz
10	October 26, 2026	Unit 4 – Manufacturing Processes: How Parts Are Made and What To Check
11	November 2, 2026	Unit 4 (Continued)
12	November 9, 2026	Unit 4 (Continued)
13	November 16, 2026	Unit 4 (Continued) Unit 4 Quiz
14	November 23, 2026	Unit 5 – Keeping Metal in Service: Welding, Failure, Corrosion, and Modern Technology
15	November 30, 2026	Unit 5 (Continued)
16	December 7, 2026	Unit 5 (Continued) Unit 5 Quiz
17	December 18, 2026	Final Exam

*Tentative schedule that may change during semester. Updates will be provided by instructor.